RESEARCH COMPILATION 2026

Mathematics of Learning

Information theory, compression, learning theory, scaling laws, emergence and grokking: why learning works, in 47 landmark papers.

Intelligence Research Atlas - Volume 01 of 6 -

Compiled 2026-10-25

Table of Contents

Mathematical Foundations of Intelligence 2

Part I - Distilled Findings 2

F1 - Understanding IS compression . . . 2

F2 - No learning without inductive bias . . . 2

F3 - Universality is cheap; depth is exponential . . . 2

F4 - Overparameterization helps, and classical theory was wrong about it . . . 3

F5 - Two training regimes: lazy (kernel) vs rich (feature learning) . . . 3

F6 - Good representations sit at an information bottleneck . . . 3

F7 - The loss landscape is non-convex but geometrically benign (with the right architecture) . . . 3

F8 - Scaling follows smooth power laws (until data binds) . . . 3

F9 - Emergence: thresholds, or an artifact of the yardstick? . . . 3

F10 - Grokking: networks memorize first, understand later — and you can engineer which one you get . . . 3

F11 - The same mathematical patterns recur in nature, brains, and neural networks . . . 4

F12 - Learning has phase transitions (statistical mechanics of intelligence) . . . 4

F13 - Intelligence has formal definitions — use them as design targets . . . 4

F14 - Energy-based views unify memory, learning, and inference . . . 4

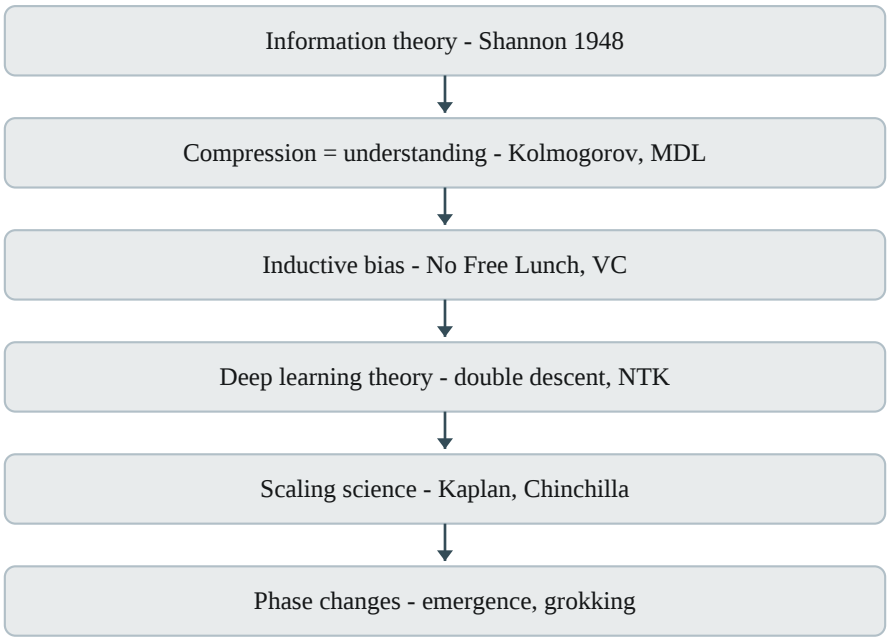
Part II - The Canon 4

Where to go next . . . 9

Mathematical Foundations of Intelligence

The mathematics that makes learning possible - and predictable. 47 landmark papers from Shannon (1948) to the 2026 limit theory of foundation models, distilled into 14 findings.

Mathematics of Learning - the through-line



Part I - Distilled Findings

F1 - Understanding IS compression

Implication for building intelligence: Every loss curve is a compression curve. Evaluating models in bits-per-character, framing pretraining as compression, and data-filtering-as-den oising all fall out of this view. If you build intelligence, you are building a compressor whether you admit it or not.

F2 - No learning without inductive bias

Implication for building intelligence: Architecture choice = hypothesis about the world. When designing a system, decide which biases you encode (locality? translation? causal order?) - that decision dominates data volume.

F3 - Universality is cheap; depth is exponential

Implication for building intelligence: Hierarchy is not a fashion - it is a complexity-theoretic necessity for compositional worlds (language, vision, planning). Shallow-but-wide is a losing strategy.

F4 - Overparameterization helps, and classical theory was wrong about it

Implication for building intelligence: Justifies training enormous models on modest data; warns that the interpolation-peak region (model size just barely fits data) is a danger zone. Also underlies why grokking (F10) exists.

F5 - Two training regimes: lazy (kernel) vs rich (feature learning)

Implication for building intelligence: If you over-initialize or over-regularize toward stability, you get a kernel machine - smooth but no feature discovery. Feature learning requires living with richer, riskier dynamics.

F6 - Good representations sit at an information bottleneck

Implication for building intelligence: Use IB as a design principle (auxiliary compression losses, VIB, representation regularization) - not as settled law. The debate itself (2017–2018) is a masterclass in how theory should interact with empirics.

F7 - The loss landscape is non-convex but geometrically benign (with the right architecture)

Implication for building intelligence: Residual streams (ResNet -> transformer) are why 100+ layer stacks train at all. If your novel architecture lacks identity-path geometry, expect optimization pain regardless of expressivity.

F8 - Scaling follows smooth power laws (until data binds)

Implication for building intelligence: Pretraining economics is now an engineering optimization problem. Compute-optimal recipes (Chinchilla) governed 2022–2024; the frontier questions are data quality/quantity (unique tokens running out) and post-training scaling (domain 03, F7/F8).

F9 - Emergence: thresholds, or an artifact of the yardstick?

Implication for building intelligence: Two lessons: (1) measure with smooth metrics or you will hallucinate capability jumps; (2) real internal phase transitions do happen - monitor mechanisms, not just benchmarks, before promising capabilities at scale N.

F10 - Grokking: networks memorize first, understand later - and you can engineer which one you get

Implication for building intelligence: Weight decay and long training are generalization instruments, not afterthoughts. If a small model "refuses" to generalize, the fix may be more training time + stronger norm control + data distribution coverage - not more parameters.

F11 - The same mathematical patterns recur in nature, brains, and neural networks

Implication for building intelligence: When choosing priors for a new architecture, steal from the patterns nature already converged on: heavy-tailed distributions, multi-scale abstraction, sparse codes, near-critical dynamics. These are the universe's default settings for complex systems.

F12 - Learning has phase transitions (statistical mechanics of intelligence)

Implication for building intelligence: Expect discontinuities when scaling ANY learning system - smooth averages hide sharp per-run transitions. Budget experiments around thresholds (capacity, data coverage, initialization scale).

F13 - Intelligence has formal definitions - use them as design targets

Implication for building intelligence: If you optimize skill (benchmarks), you get skill, not intelligence. Decide explicitly which definition your system targets; design for sample efficiency and transfer if you target Chollet-intelligence.

F14 - Energy-based views unify memory, learning, and inference

Implication for building intelligence: If you are designing new intelligence substrate, energy-based formulation gives you memory + inference + learning under one objective - a path around backprop's biological implausibility (domain 04, F6).

Part II - The Canon

47 landmark papers in chronological order. Full entries with links and keywords live in the corpus folder 01_mathematical-foundations/papers.md.

Year	Paper / Authors	Core idea	Why it matters
1936	**On Computable Numbers, with an Application to the Entscheidungsproblem** Alan Turing	Defines the Turing machine - the minimal formal model of computation - and proves some problems are undecidable.	Establishes the ceiling of what any intelligence, biological or artificial, can compute; every result below inherits this frame.
1943	**A Logical Calculus of the Ideas Immanent in Nervous Activity** Warren McCulloch, Walter Pitts	Neurons as threshold logic units; networks of them compute any logical proposition.	First proof that neural substrate => computation; the founding bridge between brain and mathematics.
1948	**A Mathematical Theory of Communication** Claude Shannon	Information = reduction of uncertainty (entropy); source coding and channel coding theorems; the bit.	The vocabulary of every loss function (cross-entropy = Shannon information) and of compression-as-understanding. The single most-used equation in LLM training.

Year	Paper / Authors	Core idea	Why it matters
1964	**A Formal Theory of Inductive Inference** Ray Solomonoff	Predict the future by Bayesian mixture over all computable hypotheses, weighted by 2^(−description length) (algorithmic probability).	The mathematically ideal - though uncomputable - learner; modern sequence models are its practical approximation.
1965	**Three Approaches to the Quantitative Definition of Information** Andrey Kolmogorov	Kolmogorov complexity: the length of the shortest program that outputs an object.	"Understanding = compression" made precise; ties Occam's razor to computability.
1971	**On the Uniform Convergence of Relative Frequencies to Probabilities (VC theory)** Vladimir Vapnik, Alexey Chervonenkis	Generalization is controlled by the capacity (VC dimension) of the hypothesis class, not just sample count.	Classical generalization theory - and the benchmark modern deep learning spectacularly violated (see Zhang 2017), forcing new theory.
1978	**Modeling by Shortest Data Description** Jorma Rissanen	Minimum Description Length (MDL): prefer the model that compresses the data plus model code best.	Practical model-selection principle uniting compression and generalization; ancestor of bits-per-byte evals.
1982	**Simplified Neuron Model as a Principal Component Analyzer** Erkki Oja	A single neuron with Hebbian plasticity plus normalization converges to the first principal component of its input.	Local, biologically plausible rules can compute global statistics - dimensionality reduction for free.

Year	Paper / Authors	Core idea	Why it matters
1982	**Neural Networks and Physical Systems with Emergent Collective Computational Abilities** John Hopfield	Recurrent binary networks as energy minima = content-addressable memory; physics (spin glasses) applied to neural systems.	Founded the statistical-mechanics view of neural networks; modern "attention as associative memory" results trace here (Nobel Prize 2024, shared with Hinton).
1984	**A Theory of the Learnable** Leslie Valiant	PAC learning: learnable = efficiently learnable to arbitrary accuracy with high probability.	Put learning itself on computational footing - learning has complexity classes like computation does.
1985	**A Learning Algorithm for Boltzmann Machines** David Ackley, Geoffrey Hinton, Terrence Sejnowski	Stochastic energy-based networks trained by contrasting data and model statistics.	First usable learning rule for hidden-unit networks; ancestor of contrastive divergence, wake-sleep, and modern self-supervised losses.
1986	**Learning Representations by Back-Propagating Errors** David Rumelhart, Geoffrey Hinton, Ronald Williams	Compute gradients of error through layers via the chain rule, making internal representations trainable.	The learning algorithm of the entire deep learning era; still how every LLM trains.
1987	**Self-Organized Criticality: An Explanation of 1/f Noise** Per Bak, Chao Tang, Kurt Wiesenfeld	Driven dissipative systems self-tune to a critical point between order and chaos, producing scale-invariant avalanches.	Candidate explanation for brain criticality (domain 04) and for why edge-of-chaos dynamics keep recurring in learning systems.
1988	**The Space of Interactions in Neural Network Models (statistical mechanics of learning)** Elizabeth Gardner	Compute the volume of parameter space consistent with a training set - capacity thresholds and phase transitions in learning.	Learning has thermodynamic phase transitions; this lens predicts grokking-like phenomena decades early.
1989	**Multilayer Feedforward Networks Are Universal Approximators** Kurt Hornik, Maxwell Stinchcombe, Halbert White	One hidden layer with enough units approximates any continuous function (independently: Cybenko 1989).	Guarantees representational sufficiency - but says nothing about depth efficiency or learnability (see Poole 2017).

Year	Paper / Authors	Core idea	Why it matters
1995	**An Information-Maximization Approach to Blind Separation and Blind Deconvolution** Anthony Bell, Terrence Sejnowski	Maximize output entropy (InfoMax) and the network discovers independent sources - without labels.	Information-theoretic self-supervision works; direct ancestor of contrastive and predictive self-supervised learning.

Year	Paper / Authors	Core idea	Why it matters
1996	**The Lack of A Priori Distinctions Between Learning Algorithms (No Free Lunch)** David Wolpert	Averaged over all possible problems, no learner beats random guessing.	Generalization must come from inductive bias matched to the world's structure - explains why architecture and priors matter.
1998	**Collective Dynamics of 'Small-World' Networks** Duncan Watts, Steven Strogatz	Networks interpolating regular/random achieve high clustering with short path lengths (small-world).	The architecture of brains, and increasingly of ML systems (MoE routing, agent swarms); a recurring mathematical pattern.
1999	**Emergence of Scaling in Random Networks** Albert-László Barabási, Réka Albert	Growth + preferential attachment produces power-law degree distributions (scale-free networks).	Power laws appear in language (Zipf), brains, and model scaling - one generating mechanism, many systems.
1999	**The Information Bottleneck Method** Naftali Tishby, Fernando Pereira, William Bialek	Optimal representations maximally compress X about irrelevant detail while preserving information about relevance variable Y; formally rate-distortion with a relevance constraint.	The leading formal account of what a "good representation" is; frames deep learning as progressive information compression.
2005	**Universal Artificial Intelligence / AIXI** Marcus Hutter (book); with Shane Legg	Intelligence = expected reward maximization over all computable environments weighted by algorithmic probability (AIXI); uncomputable but defines the optimum.	The formal reference point every practical intelligence (including LLMs) is measured against; Legg-Hutter gives the field's cleanest intelligence definition.
2014	**An Exact Mapping Between the Variational Renormalization Group and Deep Learning** Pankaj Mehta, David Schwab	Stacked restricted Boltzmann machines perform the variational renormalization group exactly.	Physics' tool for scale-dependent structure (RG) ≈ deep learning's stacked abstraction; concrete support for "deep = multi-scale".
2015	**Deep Learning and the Information Bottleneck Principle** Naftali Tishby, Noga Zaslavsky	Deep networks should first fit (memorize) then compress representations - phases observed via mutual-information probes.	Predicted a two-phase training dynamic later echoing in grokking (F10).
2016	**Group Equivariant Convolutional Networks** Taco Cohen, Max Welling	Build symmetry (translation, rotation equivariance) into architecture rather than learning it from data.	Symmetry is the strongest inductive bias physics knows; equivariant nets show architectural priors can replace data volume.

Year	Paper / Authors	Core idea	Why it matters
2016	**Exponential Expressivity in Deep Neural Networks through Transient Chaos** Ben Poole, Surya Ganguli, Maithra Raghu (with "On the Expressivity of Deep Neural Networks", Raghu et al. 2017)	Depth yields exponential gains in expressivity for representing compositional functions, beyond what width can buy.	Why we stack layers instead of widening flat ones; why hierarchies (brains included) are deep.

Year	Paper / Authors	Core idea	Why it matters
2017	**Understanding Deep Learning Requires Rethinking Generalization** Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, Oriol Vinyals	Deep nets can fit random labels perfectly, so generalization cannot follow from classical capacity control alone.	Broke VC-style theory of deep learning; opened implicit-regularization and double-descent research lines.
2017	**Opening the Black Box of Deep Neural Networks via Information** Ravid Shwartz-Ziv, Naftali Tishby	Empirically observes the fit-then-compress information phases during SGD training.	Made information-plane analysis famous - and triggered the field's sharpest theory debate.
2018	**On the Information Bottleneck Theory of Deep Learning** Andrew Saxe et al.	Compression phases largely vanish with ReLU/unbounded activations and realistic datasets; IB theory is not the mechanism.	The decisive rebuttal - science working; the field now treats IB as a design principle, not a law.
2018	**Neural Tangent Kernel: Convergence and Generalization in Neural Networks** Arthur Jacot, Franck Gabriel, Clément Hongler	In the infinite-width limit, training a network equals kernel regression with a fixed kernel (NTK).	Makes one training regime exactly solvable - and marks its boundary: NTK-lazy networks do NOT learn features.
2018	**Visualizing the Loss Landscape of Neural Nets** Hao Li, Zheng Xu, Gavin Taylor, Christoph Studer, Tom Goldstein	Filter-normalized visualization shows residual nets have trainable convex-ish valleys; without skip connections landscapes are chaotic.	Explains why residual connections (ResNet, transformer) made depth trainable - geometry, not just gradients.
2019	**On Lazy Training in Differentiable Programming** Lénaïc Chizat, Edouard Oyallon, Francis Bach	Two regimes: lazy (kernel-like, features fixed) vs rich (feature learning); initialization scale selects the regime.	Feature learning - the rich regime - is what makes deep learning work; connects to grokking transitions.
2019	**Reconciling Modern Machine-Learning Practice and the Classical Bias–Variance Trade-Off** Mikhail Belkin, Daniel Hsu, Siyuan Ma, Soumik Mandal	Double descent: test error descends, peaks at the interpolation threshold, then descends again as overparameterization grows.	The modern generalization curve; explains why billion-parameter models on "small" data behave well.

Year	Paper / Authors	Core idea	Why it matters
2019	**Deep Double Descent: Where Bigger Models and More Data Hurt** Preetum Nakkiran et al.	Double descent is ubiquitous across model size, data size, and training time in modern deep learning.	Practical warning: more parameters/data can hurt before they help; scale results have regime boundaries.
2019	**On the Measure of Intelligence** François Chollet	Intelligence = skill-acquisition efficiency over novel tasks, not skill itself; introduces the ARC benchmark built on core knowledge priors.	Corrects the benchmark-culture error (skill ≠ intelligence); ARC-AGI remains the reference test for generalization (domain 05).
2020	**Neural Collapse: An Analysis of the Modes of Neural Network Classifiers** Vardan Papyan, X. Y. Han, David Donoho	At training's end, last-layer features collapse to class means arranged as a simplex ETF - a sharp geometric universal.	2025 work unifies grokking and information bottleneck via neural collapse; a concrete geometry of "converged" learning.
2020	**Scaling Laws for Neural Language Models** Jared Kaplan et al. (OpenAI)	Loss follows smooth power laws in parameters, data, and compute, across 6+ orders of magnitude.	Made scaling a predictive engineering discipline: you can read next year's loss off this year's curves.
2022	**Training Compute-Optimal Large Language Models (Chinchilla)** Jordan Hoffmann et al. (DeepMind)	For fixed compute, parameters and tokens should scale equally (~20 tokens/parameter) - most 2021-era models were badly undertrained.	The recipe behind every efficient model since (LLaMA etc.); GPT-3-sized results from 4× smaller models.

Year	Paper / Authors	Core idea	Why it matters
2022	**Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets** Alethea Power et al.	Networks memorize quickly but suddenly generalize long after overfitting, when trained past the point of zero training loss.	Separates memorization from understanding in time; a lab model of "aha moment" learning.
2022	**Towards Understanding Grokking: An Effective Theory of Representation Learning** Ziming Liu, Ouail Kitouni, Mike Nolte, Eric Michaud, Max Tegmark, Mike Williams	Grokking = delayed generalization explained by a effective-theory of representation learning dynamics.	First mechanistic theory; predicts when grokking happens from dataset properties.
2023	**Omnigrok: Grokking Beyond Algorithmic Data** Ziming Liu, Eric Michaud, Max Tegmark	Grokking occurs on image datasets too; driven by weight-norm scale - "LU mechanism": memorize large-norm, generalize small-norm.	Grokking is general, and controllable via weight decay - a knob for generalization.

Year	Paper / Authors	Core idea	Why it matters
2023	**Progress Measures for Grokking via Mechanistic Interpretability** Neel Nanda, Lawrence Chan, Tom Lieberum, Jess Smith, Jacob Steinhardt	Reverse-engineers a grokking network: it transitions from memorization to a generalizing Fourier/multiplication circuit; restricted-set "pre-grokking" progress measures predict the transition.	The cleanest causal story of emergent generalization inside a network; mechanistic interpretability's flagship result.
2023	**Scaling Data-Constrained Language Models** Niklas Muennighoff et al.	Repeating data helps up to ~4 epochs, then returns diminish sharply; unique tokens dominate value.	The data side of scaling laws - governs planning when internet text runs out; motivates synthetic data and RL data loops.
2024	**Grokking as the Transition from Lazy to Rich Training Dynamics** Kaifeng Lyu, Zhiyuan Li, et al. (ICLR 2024)	Networks grok when they escape the lazy kernel regime into rich feature learning late in training.	Unifies grokking with the NTK/lazy-rich framework - one theory for both phenomena.
2025	**Grokking Explained: A Statistical Phenomenon** (authors: see paper)	Grokking reinterpreted as a distribution shift between training and test data rather than an optimization mystery.	The 2025 theory pivot: grokking may be statistical, not dynamical - with direct implications for what "generalization" means for LLM evals.
2025	**A Survey on Grokking** (survey; ACM Computing Surveys)	Systematizes all grokking explanations: norm-based, circuit-formation, statistical, physics-based (glass relaxation).	Entry point to the whole grokking literature; includes 2025 evidence of grokking during 7B-parameter pretraining (OLMoE).
2025	**Superposition Yields Robust Neural Scaling** (NeurIPS 2025; Anthropic toy-model based)	Neural scaling laws arise mechanistically from superposition - networks storing more features than neurons in interference-prone ways.	First mechanistic account of WHY scaling laws exist, linking interpretability (domain 03) with scaling theory.
2026	**A Limit Theory of Foundation Models** (authors: see paper)	A rigorous mathematical limit theory for foundation models, extending scaling-law analysis to data quality and training strategies.	Theoretical frontier (as of 2026): scaling laws are becoming a full limit theory rather than curve fits.

Full sources with links live in the corpus: papers.md / papers.json per domain.

Where to go next

The living version of this canon lives in the corpus folder for this domain: papers.md carries every entry with full links and keywords, findings.md holds the complete distilled findings, and the diagrams folder contains Mermaid sources that render in Obsidian and GitHub. To see how this domain connects to the rest of the atlas, read the cross-domain map in 00_master-atlas; to extend the corpus, follow the conventions in AGENTS.md and re-run the index and PDF build scripts in tools/.